

Expelexia Lab - Data Analysis Report

File Name: geophone-sensor-data.csv

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Prepared For: Expelexia (AI Analytics Program)

Project: Expelexia Lab

This report provides a comprehensive analysis and AI-generated insights based on the uploaded data.

Executive Summary

This report summarizes analysis results for geophone-sensor-data.csv using the same data shown in the website dashboard.

The dataset contains 1800 records across 14 variables.

Main areas needing attention: mean: no issues, top_3_mean: 12 anomalies, min: no issues.

Observed trend signals: mean down, top_3_mean up, min down.

Recommendations focus on practical monitoring, early detection of unusual changes, and improving reliability of future data collection.

Data Overview

Measure	Value	What it means
Number of records	1800	How many entries were analyzed.
Number of variables	14	How many fields were evaluated.
Missing values	{'mean': 0, 'top_3_mean': 0, 'min': 0, 'max': 0, 'std_dev': 0, 'median': 0, 'q1': 0, 'q3': 0, 'skewness': 0, 'dominant_freq': 0, 'energy': 0}	Missing entries can reduce result reliability if frequent.

Website Dashboard Alignment

This table mirrors key trend, focus, and confidence indicators used in the website dashboard view.

Metric	Trend	Focus Area	Confidence
mean	down	no issues	1.000
top_3_mean	up	12 anomalies	0.990
min	down	no issues	1.000
max	down	18 anomalies	0.990
std_dev	down	high variance	1.000
median	down	no issues	1.000

Metric	Trend	Focus Area	Confidence
q1	up	6 anomalies	1.000
q3	down	no issues	1.000
skewness	up	16 anomalies, high variance	0.990
dominant_freq	down	high variance	1.000

Visual Analysis

The chart below shows how key values compare. It helps identify increases, decreases, and unusual patterns over time.

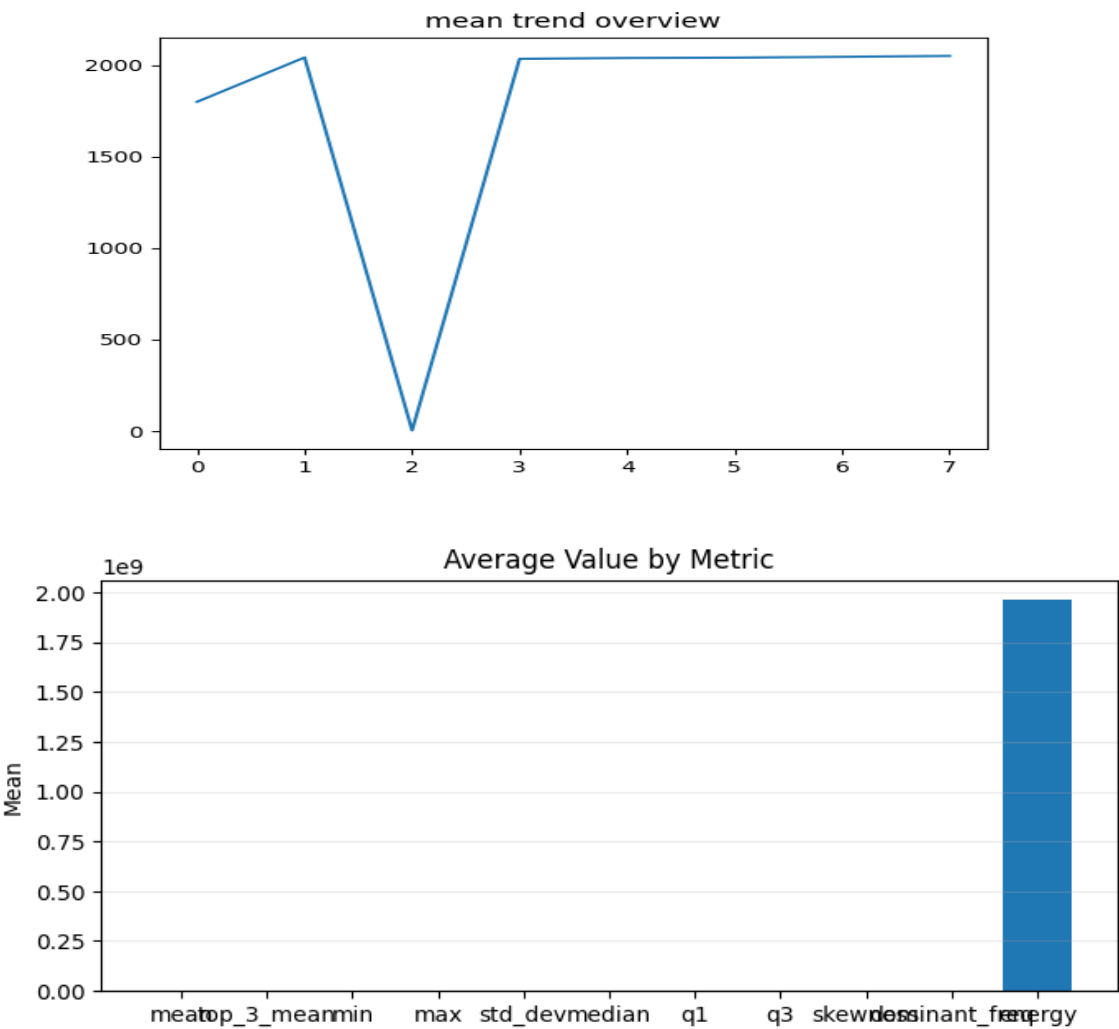


Chart Interpretation:

- mean: bar chart centers near mean 2042.247; line trend indicates down. Mean/median relation suggests a right-skewed distribution.
- top_3_mean: bar chart centers near mean 2064.690; line trend indicates up. Mean/median relation suggests a right-skewed distribution.
- min: bar chart centers near mean 2016.367; line trend indicates down. Mean/median relation suggests a left-skewed distribution.
- max: bar chart centers near mean 2067.156; line trend indicates down. Mean/median relation suggests a

right-skewed distribution.

- std_dev: bar chart centers near mean 8.529; line trend indicates down. Mean/median relation suggests a right-skewed distribution.

- median: bar chart centers near mean 2043.396; line trend indicates down. Mean/median relation suggests a right-skewed distribution.

- q1: bar chart centers near mean 2037.628; line trend indicates up. Mean/median relation suggests a right-skewed distribution.

- q3: bar chart centers near mean 2048.030; line trend indicates down. Mean/median relation suggests a right-skewed distribution.

- skewness: bar chart centers near mean -0.347; line trend indicates up. Mean/median relation suggests a left-skewed distribution.

- dominant_freq: bar chart centers near mean 410.099; line trend indicates down. Mean/median relation suggests a right-skewed distribution.

- energy: bar chart centers near mean 1961491633.142; line trend indicates down. Mean/median relation suggests a right-skewed distribution.

- Histogram bins show where value density is concentrated between quartile ranges.

- Pie chart summarizes stable vs attention-needed metrics from focus-area rules.

Data Change Explanation

Data Change Analysis:

- mean: value range 2034.811 to 2050.709 (spread 15.898), middle 50% spans 6.340. Trend is down; volatility appears moderate; focus area is no issues.

- top_3_mean: value range 2039.000 to 2232.667 (spread 193.667), middle 50% spans 18.000. Trend is up; volatility appears moderate; focus area is 12 anomalies.

- min: value range 1948.000 to 2045.000 (spread 97.000), middle 50% spans 23.000. Trend is down; volatility appears moderate; focus area is no issues.

- max: value range 2039.000 to 2261.000 (spread 222.000), middle 50% spans 22.000. Trend is down; volatility appears moderate; focus area is 18 anomalies.

- std_dev: value range 1.375 to 37.679 (spread 36.304), middle 50% spans 1.997. Trend is down; volatility appears high; focus area is high variance.

- median: value range 2034.000 to 2056.000 (spread 22.000), middle 50% spans 8.000. Trend is down; volatility appears moderate; focus area is no issues.

- q1: value range 2021.000 to 2049.000 (spread 28.000), middle 50% spans 5.000. Trend is up; volatility appears moderate; focus area is 6 anomalies.

- q3: value range 2037.000 to 2069.000 (spread 32.000), middle 50% spans 8.000. Trend is down; volatility appears moderate; focus area is no issues.

- skewness: value range -4.396 to 1.639 (spread 6.036), middle 50% spans 0.613. Trend is up; volatility appears high; focus area is 16 anomalies, high variance.

- dominant_freq: value range 1.000 to 601.000 (spread 600.000), middle 50% spans 300.000. Trend is down; volatility appears high; focus area is high variance.

- energy: value range 1915725274.000 to 2013152581.000 (spread 97427307.000), middle 50% spans 39813566.500. Trend is down; volatility appears moderate; focus area is no issues.

Key Findings

- mean: values range from 2034.81 to 2050.71, average around 2042.25. Trend appears down; review note: no issues.

- top_3_mean: values range from 2039.00 to 2232.67, average around 2064.69. Trend appears up; review note: 12 anomalies.
- min: values range from 1948.00 to 2045.00, average around 2016.37. Trend appears down; review note: no issues.
- max: values range from 2039.00 to 2261.00, average around 2067.16. Trend appears down; review note: 18 anomalies.
- std_dev: values range from 1.37 to 37.68, average around 8.53. Trend appears down; review note: high variance.
- median: values range from 2034.00 to 2056.00, average around 2043.40. Trend appears down; review note: no issues.

AI-Powered Insights

Executive Summary: This report explains the analysis results for geophone-sensor-data.csv in clear language. The findings are based on the same data used in the website dashboard and do not introduce different calculations.

Data Overview: The dataset includes approximately 1800 records and 14 variables. The analysis focused on trend behavior, consistency, unusual values, and confidence indicators.

Key Findings: Main focus areas include mean:no issues, top_3_mean:12 anomalies, min:no issues, max:18 anomalies, std_dev:high variance, median:no issues, q1:6 anomalies, q3:no issues, skewness:16 anomalies, high variance, dominant_freq:high variance, energy:no issues. Trend signals show mean:down, top_3_mean:up, min:down, max:down, std_dev:down, median:down, q1:up, q3:down, skewness:up, dominant_freq:down, energy:down.

AI-Powered Insights: The observed changes may reflect shifts in operating conditions, collection methods, or natural process variation. Results should be reviewed with context from the experiment timeline.

Confidence & Reliability: The analysis is generally reliable for decision support, but confidence can be reduced when unusual values or missing entries are present.

Technical Review (Advanced)

Technical Review (Advanced):

- mean: mean=2042.2465, std=3.6591, min=2034.8107, max=2050.7087, IQR=6.3403, CV=0.0018, trend=down, focus=no issues, confidence=1.0.
- top_3_mean: mean=2064.6896, std=28.2936, min=2039.0000, max=2232.6667, IQR=18.0000, CV=0.0137, trend=up, focus=12 anomalies, confidence=0.99.
- min: mean=2016.3672, std=28.1343, min=1948.0000, max=2045.0000, IQR=23.0000, CV=0.0140, trend=down, focus=no issues, confidence=1.0.
- max: mean=2067.1556, std=31.8005, min=2039.0000, max=2261.0000, IQR=22.0000, CV=0.0154, trend=down, focus=18 anomalies, confidence=0.99.
- std_dev: mean=8.5288, std=10.7606, min=1.3749, max=37.6790, IQR=1.9965, CV=1.2617, trend=down, focus=high variance, confidence=1.0.
- median: mean=2043.3956, std=5.0443, min=2034.0000, max=2056.0000, IQR=8.0000, CV=0.0025, trend=down, focus=no issues, confidence=1.0.
- q1: mean=2037.6283, std=4.4901, min=2021.0000, max=2049.0000, IQR=5.0000, CV=0.0022, trend=up, focus=6 anomalies, confidence=1.0.
- q3: mean=2048.0297, std=9.0186, min=2037.0000, max=2069.0000, IQR=8.0000, CV=0.0044, trend=down, focus=no issues, confidence=1.0.
- skewness: mean=-0.3473, std=0.4120, min=-4.3962, max=1.6395, IQR=0.6129, CV=1.1861, trend=up, focus=16

anomalies, high variance, confidence=0.99.

- dominant_freq: mean=410.0994, std=152.8752, min=1.0000, max=601.0000, IQR=300.0000, CV=0.3728, trend=down, focus=high variance, confidence=1.0.

- energy: mean=1961491633.1422, std=22704914.9478, min=1915725274.0000, max=2013152581.0000, IQR=39813566.5000, CV=0.0116, trend=down, focus=no issues, confidence=1.0.

Technical Recommendations (Advanced)

Technical Recommendations (Advanced):

- mean: persistent downward trend; validate process degradation assumptions and lower tolerance thresholds.
- top_3_mean: anomaly-heavy profile detected; run outlier root-cause analysis against raw capture logs.
- top_3_mean: persistent upward trend; evaluate drift alarms and upper tolerance boundaries.
- min: persistent downward trend; validate process degradation assumptions and lower tolerance thresholds.
- max: anomaly-heavy profile detected; run outlier root-cause analysis against raw capture logs.
- max: persistent downward trend; validate process degradation assumptions and lower tolerance thresholds.
- std_dev: high coefficient of variation (CV=1.2617); apply robust thresholding and recalibration checks.
- std_dev: persistent downward trend; validate process degradation assumptions and lower tolerance thresholds.
- median: persistent downward trend; validate process degradation assumptions and lower tolerance thresholds.
- q1: anomaly-heavy profile detected; run outlier root-cause analysis against raw capture logs.
- q1: persistent upward trend; evaluate drift alarms and upper tolerance boundaries.
- q3: persistent downward trend; validate process degradation assumptions and lower tolerance thresholds.

Special AI Recommendations

Priority actions below are generated from trend volatility, confidence, and focus-area signals.

Abstract: The analysis processed 1800 rows across 14 columns to evaluate quality, trend behavior, and operational risk.

Introduction: Recommendations are mapped to Performance (25%), Innovation (25%), Operational Impact (25%), and Responsible AI (25%).

Materials and Methods: Data quality, distribution spread, quartiles, trend direction, focus-area tags, and confidence values were evaluated per metric.

Results:

- mean: mean=2042.2465, std=3.6591, range=2034.8107..2050.7087, IQR=6.3403, CV=0.002, trend=down, focus=no issues, confidence=1.0.
- top_3_mean: mean=2064.6896, std=28.2936, range=2039.0000..2232.6667, IQR=18.0000, CV=0.014, trend=up, focus=12 anomalies, confidence=0.99.
- min: mean=2016.3672, std=28.1343, range=1948.0000..2045.0000, IQR=23.0000, CV=0.014, trend=down, focus=no issues, confidence=1.0.
- max: mean=2067.1556, std=31.8005, range=2039.0000..2261.0000, IQR=22.0000, CV=0.015, trend=down, focus=18 anomalies, confidence=0.99.
- std_dev: mean=8.5288, std=10.7606, range=1.3749..37.6790, IQR=1.9965, CV=1.262, trend=down, focus=high variance, confidence=1.0.
- median: mean=2043.3956, std=5.0443, range=2034.0000..2056.0000, IQR=8.0000, CV=0.002, trend=down, focus=no issues, confidence=1.0.
- q1: mean=2037.6283, std=4.4901, range=2021.0000..2049.0000, IQR=5.0000, CV=0.002, trend=up, focus=6 anomalies, confidence=1.0.
- q3: mean=2048.0297, std=9.0186, range=2037.0000..2069.0000, IQR=8.0000, CV=0.004, trend=down, focus=no issues, confidence=1.0.
- skewness: mean=-0.3473, std=0.4120, range=-4.3962..1.6395, IQR=0.6129, CV=1.186, trend=up, focus=16

anomalies, high variance, confidence=0.99.

- dominant_freq: mean=410.0994, std=152.8752, range=1.0000..601.0000, IQR=300.0000, CV=0.373, trend=down, focus=high variance, confidence=1.0.

- energy: mean=1961491633.1422, std=22704914.9478, range=1915725274.0000..2013152581.0000, IQR=39813566.5000, CV=0.012, trend=down, focus=no issues, confidence=1.0.

Discussion:

- Risk stratification: high risk=3, medium risk=3, low-confidence=0 metrics.

- High-priority metrics: std_dev, skewness, dominant_freq.

- Medium-priority metrics: top_3_mean, max, q1.

- Recommendation confidence improves when calibration records, sampling context, and anomaly root-cause notes are attached to each run.

Conclusion: The dataset supports decision guidance when recommendations are executed with metric-level evidence, uncertainty disclosure, and safety checks.

Recommended Next Actions (criteria-aligned):

1) [Performance 25%] mean: downward trend detected; verify if decline matches expected process behavior and acceptable bounds.

2) [Performance 25%] top_3_mean: anomaly concentration detected; verify raw acquisition logs, sensor drift, and boundary thresholds.

3) [Performance 25%] top_3_mean: upward trend detected; review upper tolerance limits and likely drift drivers.

4) [Performance 25%] min: downward trend detected; verify if decline matches expected process behavior and acceptable bounds.

5) [Performance 25%] max: anomaly concentration detected; verify raw acquisition logs, sensor drift, and boundary thresholds.

6) [Performance 25%] max: downward trend detected; verify if decline matches expected process behavior and acceptable bounds.

7) [Innovation 25%] std_dev: high variability (CV=1.262); add adaptive control limits and confidence-banded recommendations.

8) [Performance 25%] std_dev: downward trend detected; verify if decline matches expected process behavior and acceptable bounds.

9) [Performance 25%] median: downward trend detected; verify if decline matches expected process behavior and acceptable bounds.

10) [Performance 25%] q1: anomaly concentration detected; verify raw acquisition logs, sensor drift, and boundary thresholds.

11) [Operational Impact 25%] Log metric-level evidence, confidence tags, and safety outcomes in a consistent audit trail for transparent reporting.

Recommendations

Detailed Recommendations:

Priority 1 - Immediate Monitoring:

- top_3_mean: focus=12 anomalies, trend=up, CV=0.014, confidence=0.99. Prioritize calibration, anomaly review, and completeness checks before downstream decisions.

- max: focus=18 anomalies, trend=down, CV=0.015, confidence=0.99. Prioritize calibration, anomaly review, and completeness checks before downstream decisions.

- std_dev: focus=high variance, trend=down, CV=1.262, confidence=1.0. Prioritize calibration, anomaly review, and completeness checks before downstream decisions.

- q1: focus=6 anomalies, trend=up, CV=0.002, confidence=1.0. Prioritize calibration, anomaly review, and completeness checks before downstream decisions.

- skewness: focus=16 anomalies, high variance, trend=up, CV=1.186, confidence=0.99. Prioritize calibration, anomaly review, and completeness checks before downstream decisions.



- dominant_freq: focus=high variance, trend=down, CV=0.373, confidence=1.0. Prioritize calibration, anomaly review, and completeness checks before downstream decisions.
- Priority 2 - Investigation Actions:
- Compare unusual spikes and drops against experiment logs, equipment events, and environment changes from the same time window.
 - Validate whether trend changes are expected process behavior or indicate drift needing correction.
- Priority 3 - Improvement Actions:
- Standardize data collection checkpoints (calibration, timestamp validation, completeness checks) before each run.
 - Add weekly review of confidence and focus-area flags to improve reliability over time.
- Priority 4 - Governance and Safety:
- Require human approval for decisions when confidence is low or quality warnings are present.
 - Keep a decision log linking each action to the supporting metric evidence and confidence value.

Priority Recommendation Guide

Priority Level	Meaning	Recommended Action
High Priority	Immediate attention required	Priority 1 - Immediate Monitoring:
Moderate Attention	Track closely and follow up	Priority 2 - Investigation Actions:
Low Concern	Routine monitoring is sufficient	Priority 3 - Improvement Actions:

Confidence & Reliability

The analysis is generally reliable when interpreted with the same context shown in the website dashboard. Attention is recommended for: top_3_mean, max, std_dev, q1, skewness, dominant_freq due to unusual patterns or data quality concerns. Missing values or unusual spikes can reduce certainty, so regular monitoring and data quality checks are advised.

Document Notes

- Document Context:
- Source file: geophone-sensor-data.csv
 - File type: csv
 - Structural profile: rows=1800, columns=14
 - This report is generated for a general lab-review workflow and operational decision support.

Legal & Compliance Context

- The report automatically adapts reference guidance to detected data context. Domain tags: personal-data.
- Apply recommendations with human oversight and retain an auditable decision log linking actions to evidence.
 - Confirm data minimization, access control, and retention limits before operational use.
 - Personal-data indicators detected: apply lawful basis checks, purpose limitation, and subject-rights handling workflows.
 - Use de-identification or pseudonymization where feasible before sharing outputs.



AI Responsibility & Transparency

This report is generated using AI technologies provided through Microsoft Azure. Recommendations are based on available data and are intended to support decision-making. Validate critical decisions with domain experts.

Appendix

Additional summary details are provided below for reference.

Metric	Mean	Std	Min	Max	Trend	Focus Area	Confidence
mean	2042.2465	3.6591	2034.8107	2050.7087	down	no issues	1.0
top_3_mean	2064.6896	28.2936	2039.0000	2232.6667	up	12 anomalies	0.99
min	2016.3672	28.1343	1948.0000	2045.0000	down	no issues	1.0
max	2067.1556	31.8005	2039.0000	2261.0000	down	18 anomalies	0.99
std_dev	8.5288	10.7606	1.3749	37.6790	down	high variance	1.0
median	2043.3956	5.0443	2034.0000	2056.0000	down	no issues	1.0
q1	2037.6283	4.4901	2021.0000	2049.0000	up	6 anomalies	1.0
q3	2048.0297	9.0186	2037.0000	2069.0000	down	no issues	1.0
skewness	-0.3473	0.4120	-4.3962	1.6395	up	16 anomalies, high variance	0.99
dominant_freq	410.0994	152.8752	1.0000	601.0000	down	high variance	1.0
energy	1961491633.1422	22704914.9478	1915725274.0000	2013152581.0000	down	no issues	1.0

Detailed Data Table

Metric	Count	Mean	Std	Min	Max
mean	1800.0	2042.2465	3.6591	2034.8107	2050.7087
top_3_mean	1800.0	2064.6896	28.2936	2039.0	2232.6667
min	1800.0	2016.3672	28.1343	1948.0	2045.0
max	1800.0	2067.1556	31.8005	2039.0	2261.0
std_dev	1800.0	8.5288	10.7606	1.3749	37.679
median	1800.0	2043.3956	5.0443	2034.0	2056.0
q1	1800.0	2037.6283	4.4901	2021.0	2049.0
q3	1800.0	2048.0297	9.0186	2037.0	2069.0
skewness	1800.0	-0.3473	0.412	-4.3962	1.6395
dominant_freq	1800.0	410.0994	152.8752	1.0	601.0
energy	1800.0	1961491633.1422	22704914.9478	1915725274.0	2013152581.0

References (AI Use)

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